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(54) ELECTRONIC APPARATUS, IMAGE PROCESSING METHOD, AND STORAGE MEDIUM

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(57) ABSTRACT

An electronic apparatus comprises: a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting; a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected; a determination unit that determines a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject detected by the first detection means; an adjustment unit that adjusts the reliability based on a difference between information about a distance to each of the subjects in a depth direction and information about a distance to the object in a depth direction; and a decision unit that decides, based on the reliability adjusted by the adjustment unit, one of the subject/subjects detected by the first detecting unit as a main subject.

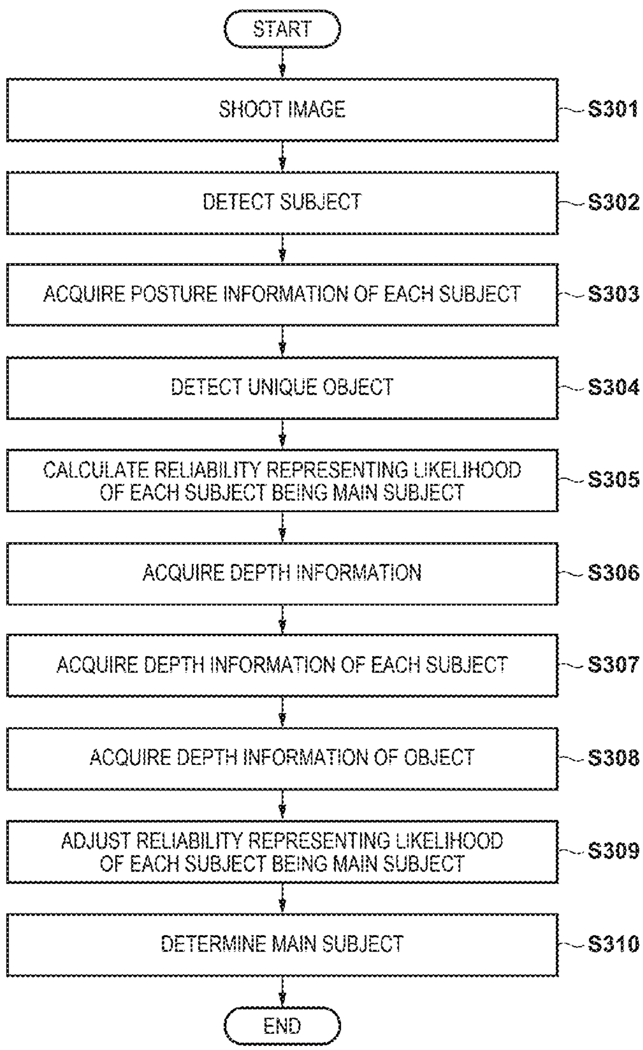


FIG. 1

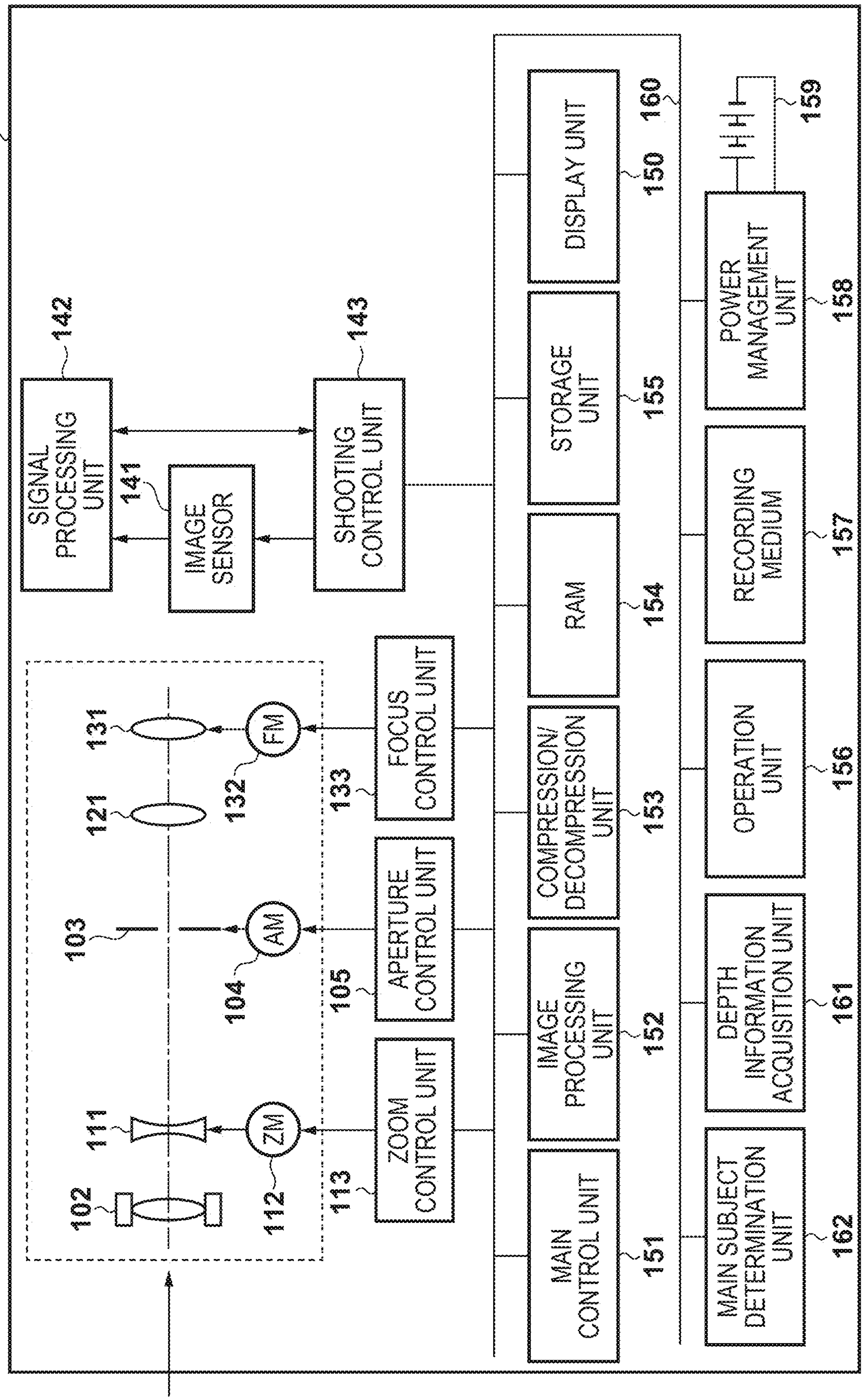

100
102
111
112
ZM
113
ZOOM CONTROL UNIT
103
104
AM
105
APERTURE CONTROL UNIT
121
131
132
FM
133
FOCUS CONTROL UNIT
142
SIGNAL PROCESSING UNIT
141
IMAGE SENSOR
143
SHOOTING CONTROL UNIT
160
MAIN CONTROL UNIT
151
IMAGE PROCESSING UNIT
152
COMPRESSION/ DECOMPRESSION UNIT
153
RAM
154
STORAGE UNIT
155
DISPLAY UNIT
150
MAIN SUBJECT DETERMINATION UNIT
162
DEPTH INFORMATION ACQUISITION UNIT
161
OPERATION UNIT
156
RECORDING MEDIUM
157
POWER MANAGEMENT UNIT
158
159

FIG. 2

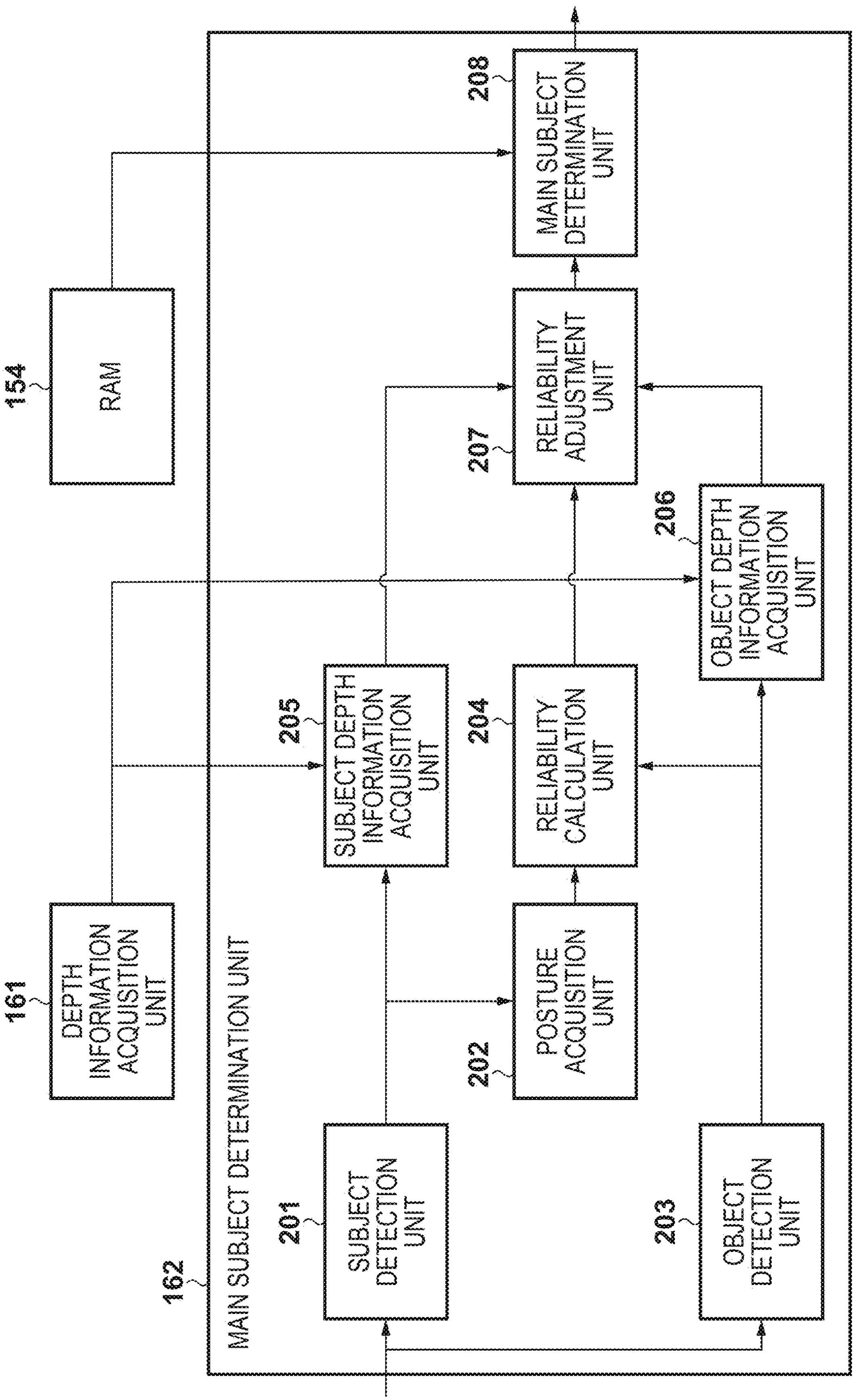

162
MAIN SUBJECT DETERMINATION UNIT
161
DEPTH INFORMATION ACQUISITION UNIT
154
RAM
201
SUBJECT DETECTION UNIT
202
POSTURE ACQUISITION UNIT
203
OBJECT DETECTION UNIT
204
RELIABILITY CALCULATION UNIT
205
SUBJECT DEPTH INFORMATION ACQUISITION UNIT
206
OBJECT DEPTH INFORMATION ACQUISITION UNIT
207
RELIABILITY ADJUSTMENT UNIT
208
MAIN SUBJECT DETERMINATION UNIT

FIG. 3

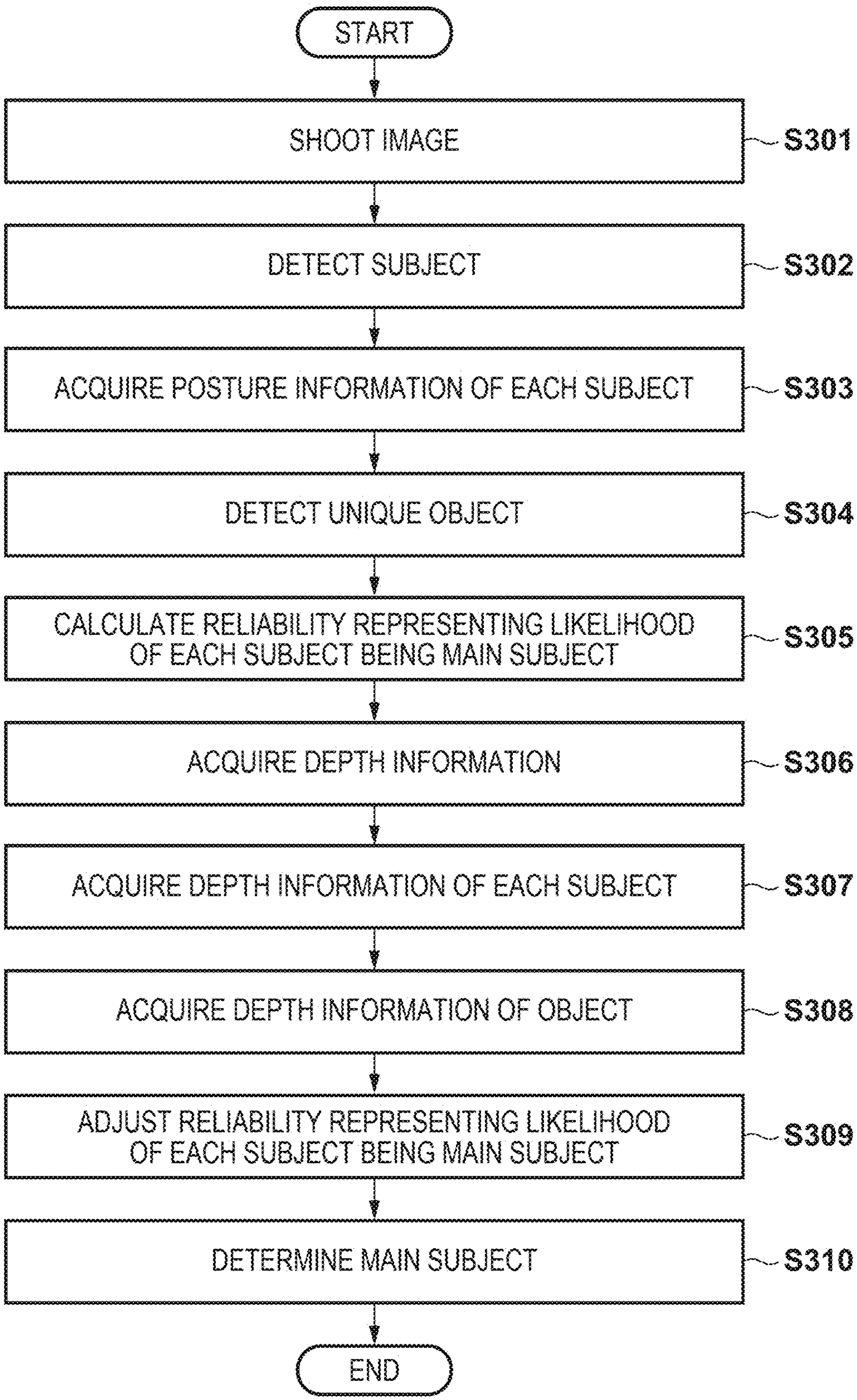

START
SHOOT IMAGE
S301
DETECT SUBJECT
S302
ACQUIRE POSTURE INFORMATION OF EACH SUBJECT
S303
DETECT UNIQUE OBJECT
S304
CALCULATE RELIABILITY REPRESENTING LIKELIHOOD OF EACH SUBJECT BEING MAIN SUBJECT
S305
ACQUIRE DEPTH INFORMATION
S306
ACQUIRE DEPTH INFORMATION OF EACH SUBJECT
S307
ACQUIRE DEPTH INFORMATION OF OBJECT
S308
ADJUST RELIABILITY REPRESENTING LIKELIHOOD OF EACH SUBJECT BEING MAIN SUBJECT
S309
DETERMINE MAIN SUBJECT
S310
END

FIG. 4A

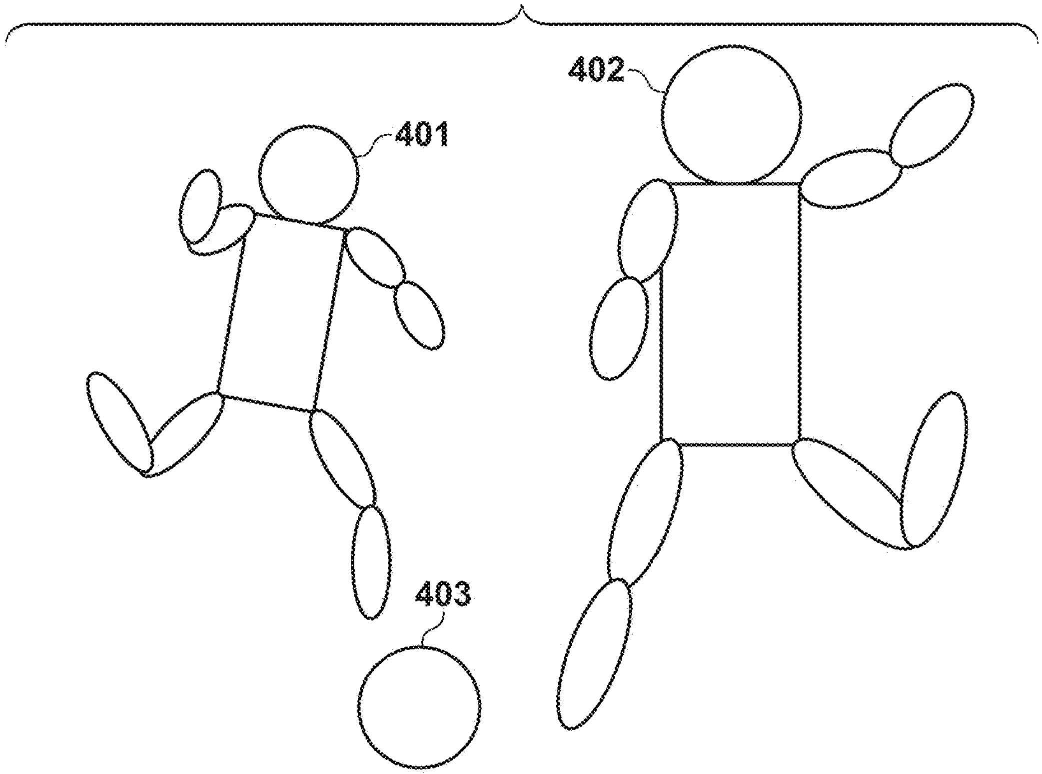


FIG. 4B

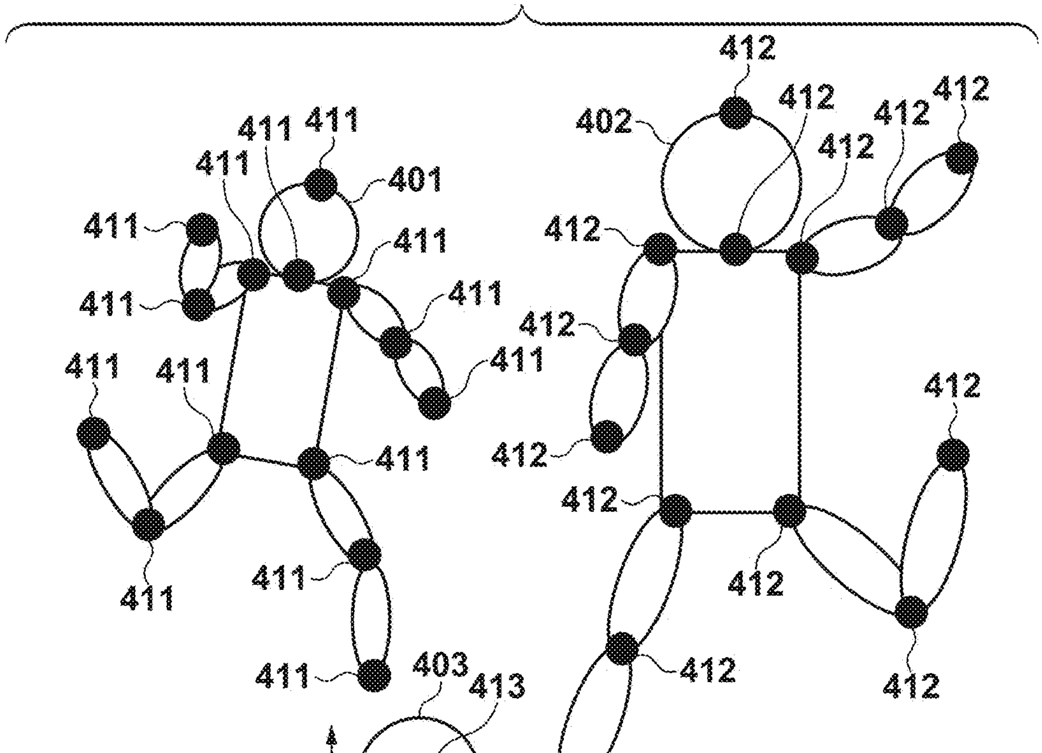


FIG. 5

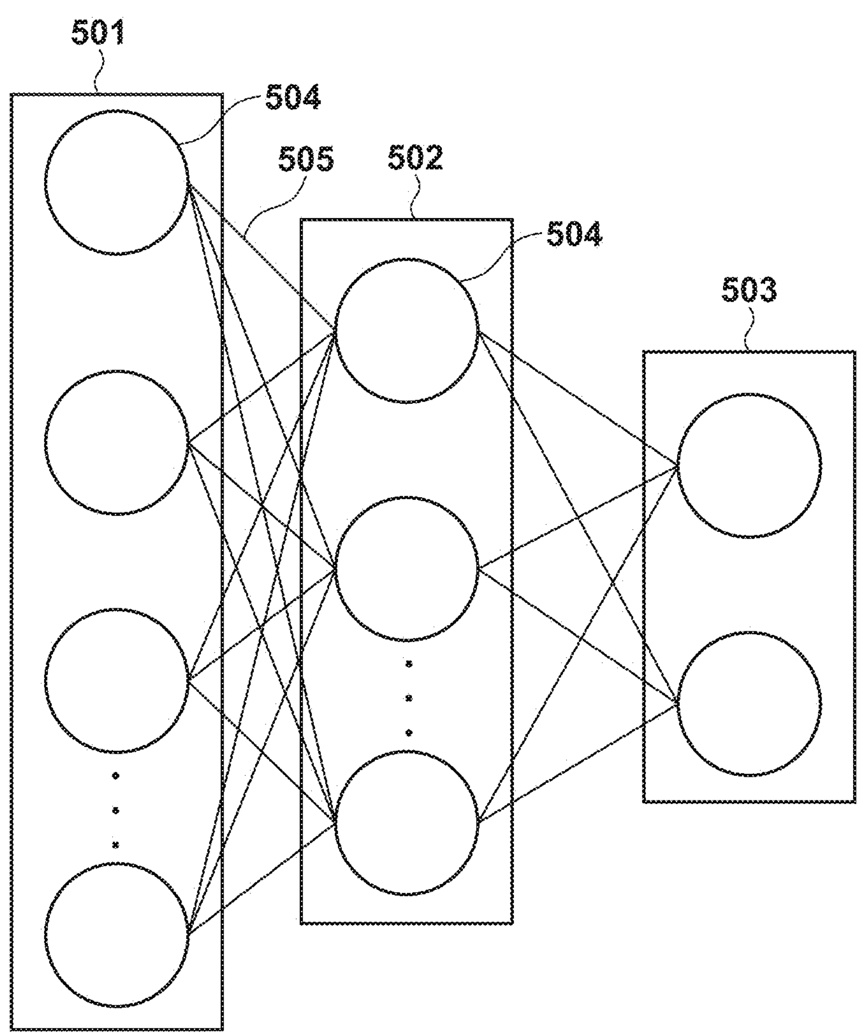


FIG. 6

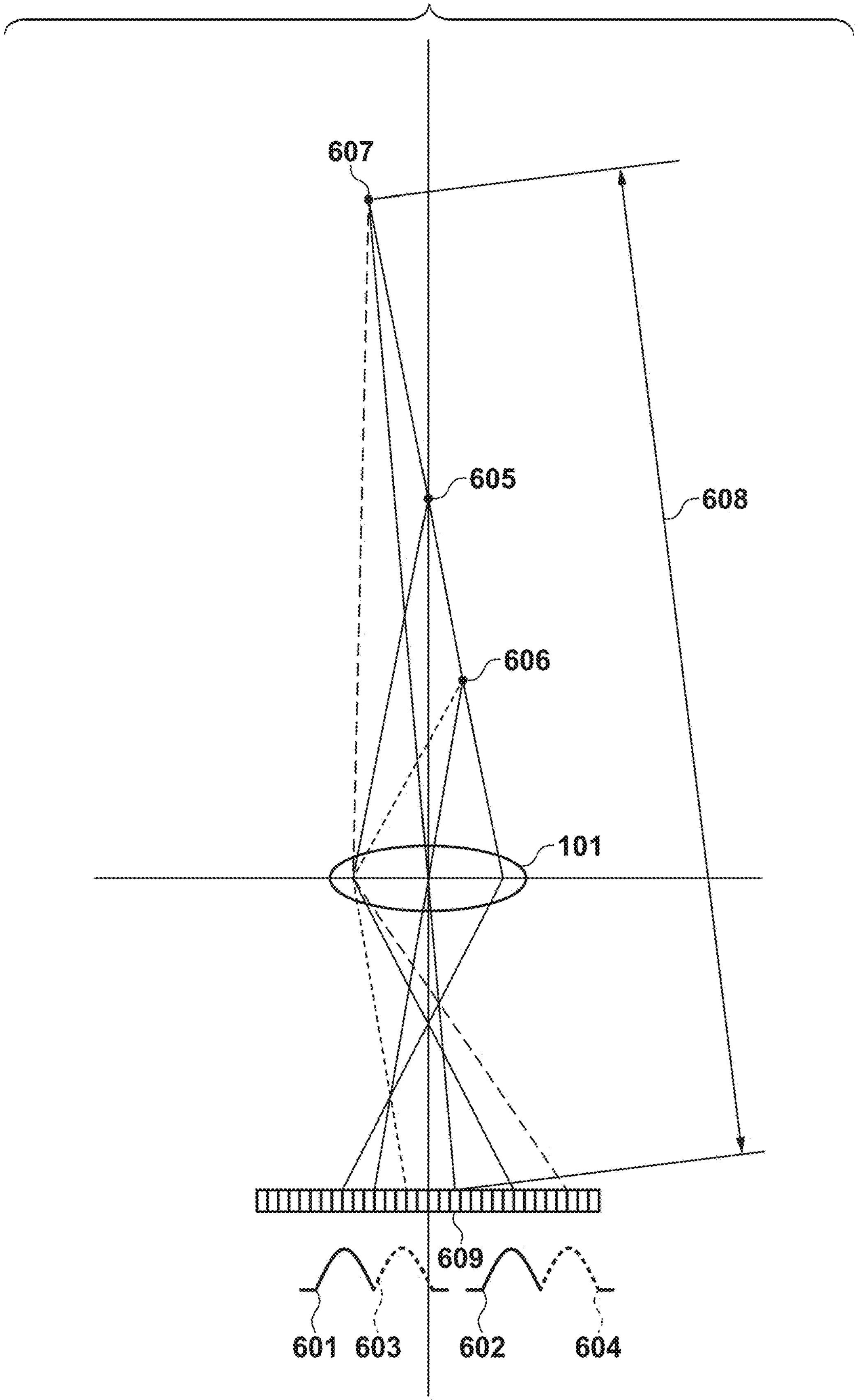


FIG. 7

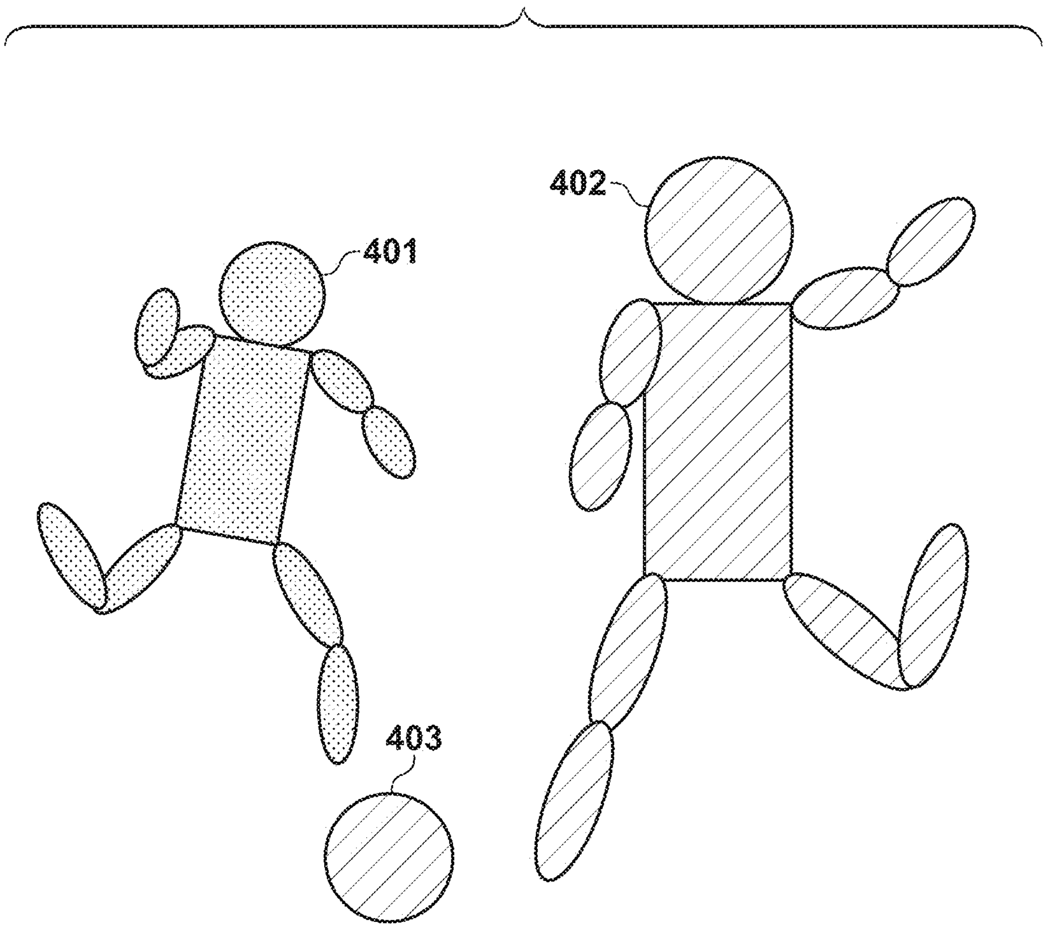


FIG. 8

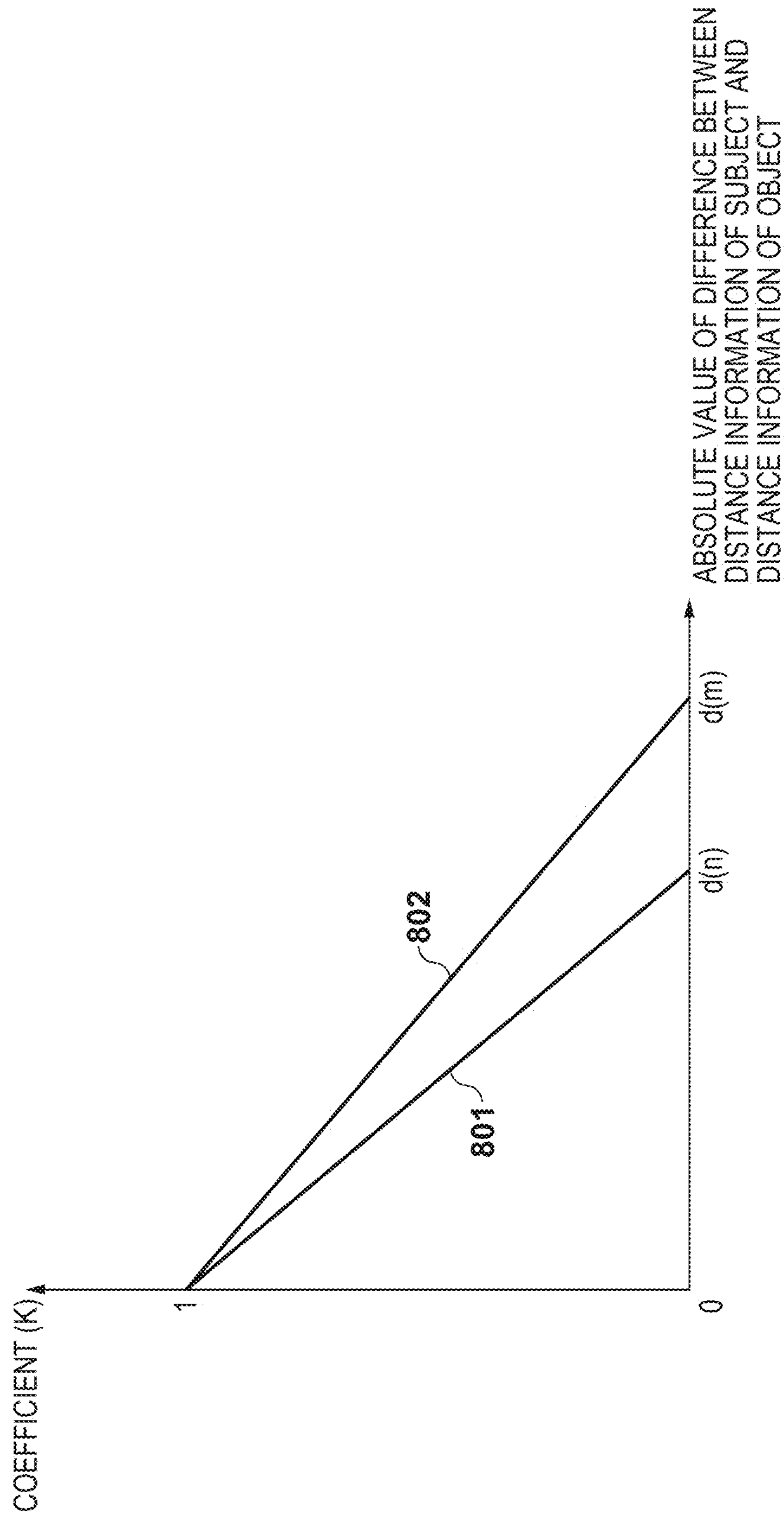

COEFFICIENT (K)
1
0
802
801
d(n)
d(m)
ABSOLUTE VALUE OF DIFFERENCE BETWEEN DISTANCE INFORMATION OF SUBJECT AND DISTANCE INFORMATION OF OBJECT

FIG. 9

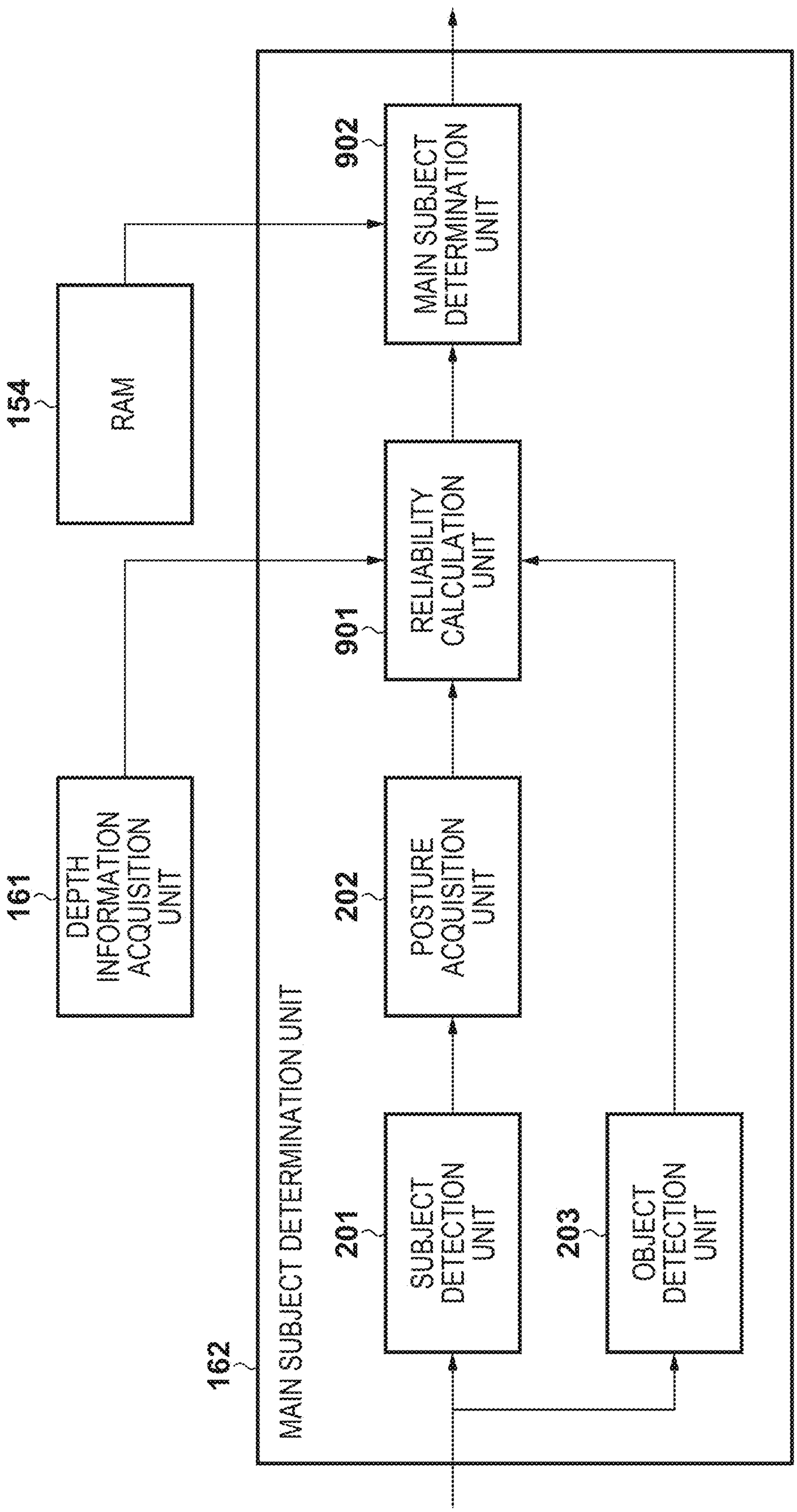
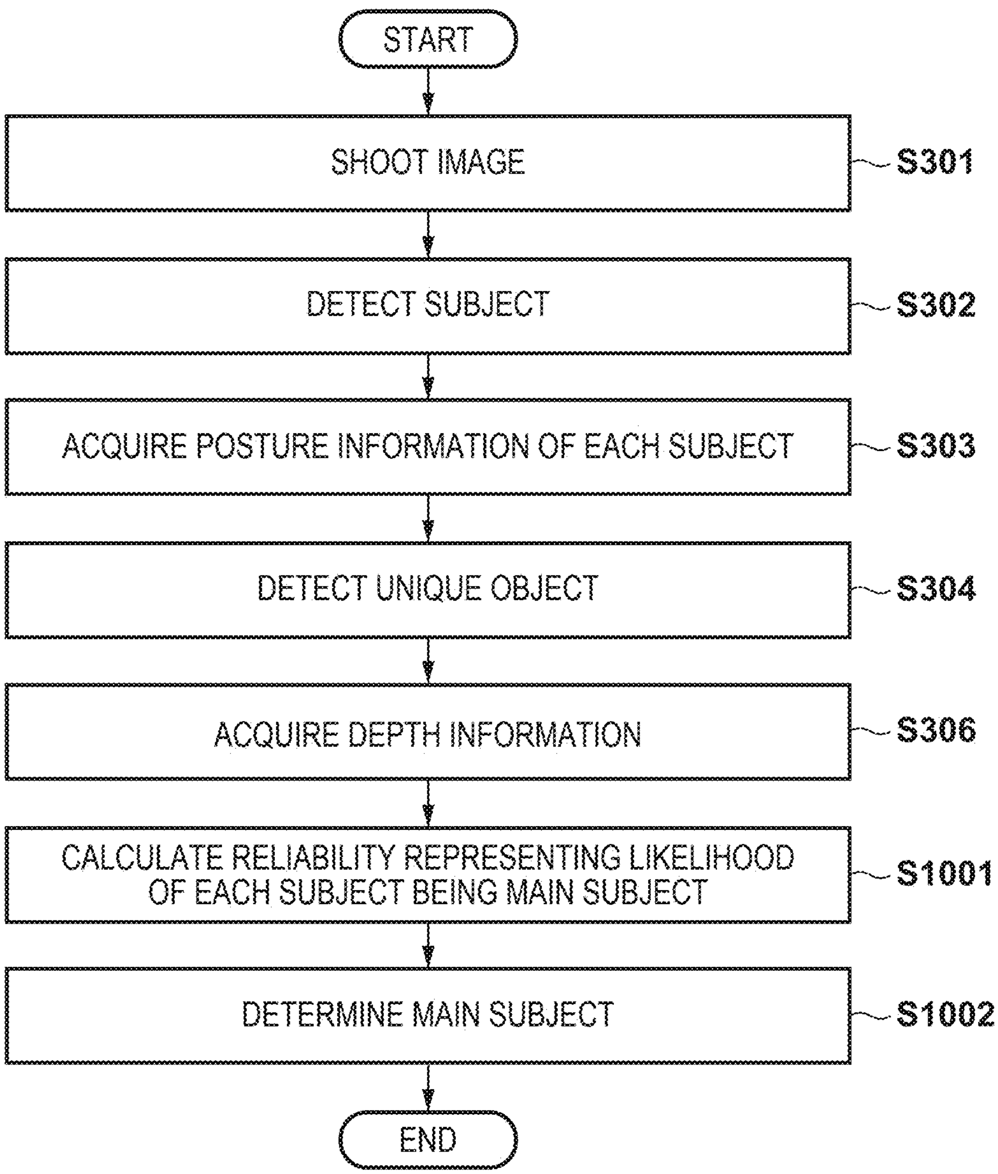

162
MAIN SUBJECT DETERMINATION UNIT
201
SUBJECT DETECTION UNIT
203
OBJECT DETECTION UNIT
202
POSTURE ACQUISITION UNIT
161
DEPTH INFORMATION ACQUISITION UNIT
901
RELIABILITY CALCULATION UNIT
154
RAM
902
MAIN SUBJECT DETERMINATION UNIT

FIG. 10


START
SHOOT IMAGE
S301
DETECT SUBJECT
S302
ACQUIRE POSTURE INFORMATION OF EACH SUBJECT
S303
DETECT UNIQUE OBJECT
S304
ACQUIRE DEPTH INFORMATION
S306
CALCULATE RELIABILITY REPRESENTING LIKELIHOOD OF EACH SUBJECT BEING MAIN SUBJECT
S1001
DETERMINE MAIN SUBJECT
S1002
END

ELECTRONIC APPARATUS, IMAGE PROCESSING METHOD, AND STORAGE MEDIUM

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an electronic apparatus, an image processing method, and a storage medium, and more particularly to a technique for determining a main subject in a captured image.

Description of the Related Art

[0002] Conventionally, in continuous shooting in which a plurality of shots are taken in succession or in moving image shooting, when shooting a plurality of detected moving subjects by an image capturing apparatus, a main subject is determined from the plurality of subjects, and the determined main subject is kept in focus. As a method for determining the main subject, Japanese Patent Laid-Open No. 2018-66889 discloses a method for detecting a subject (object) and a face to be tracked, setting the main subject based on the position and movement of the tracking target and the position of the face, and adjusting the focus on the set main subject.

[0003] However, in the method disclosed in Japanese Patent Laid-Open No. 2018-66889, a subject whose face has not been detected cannot be set as the main subject. Therefore, for example, if a subject intended by a photographer is facing backward, there is a possibility that an unintended subject will be determined to be the main subject and continue to be focused on.

SUMMARY OF THE INVENTION

[0004] The present invention has been made in consideration of the above situation, and in a case where a plurality of different subjects are detected, it is possible to determine the subject that is closest to the photographer's intention as the main subject.

[0005] According to the present invention, provided is an electronic apparatus comprising one or more processors and/or circuitry which function as: a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting; a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected; a determination unit that determines a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject detected by the first detection unit; an adjustment unit that adjusts the reliability based on a difference between information about a distance to each of the subjects in a depth direction and information about a distance to the object in a depth direction; and a decision unit that decides, based on the reliability adjusted by the adjustment unit, one of the subject/subjects detected by the first detecting unit as a main subject.

[0006] Further, according to the present invention, provided is an electronic apparatus comprising one or more processors and/or circuitry which function as: a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting; a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected; a decision unit that decides one of the subject/subjects detected by the first detection unit as a main subject based on a posture of each subject detected by the first detection unit and a difference between information about a distance to each subject in the depth direction and information about a distance to the object in the depth direction.

[0007] Furthermore, according to the present invention, provided is an electronic apparatus comprising one or more processors and/or circuitry which function as: a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting; a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected; an acquisition unit that acquires information regarding a distance in the depth direction to each subject detected by the first detection unit and information about a distance in the depth direction to the object; a determination unit that uses machine learning to determine a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject; and a decision unit that decides one of the subject/subjects detected by the first detection unit as a main subject based on the reliability, wherein the determination unit uses training data optimized by using information about the distance to the main subject previously acquired by the acquisition unit in the machine learning.

[0008] Further, according to the present invention, provided is an image processing method comprising: detecting one or more subjects from any of images obtained by repeatedly performing shooting; detecting a predetermined object from the image from which the subject/subjects are detected; determining a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each detected subject; adjusting the reliability based on a difference between information about a distance to each of the subjects in a depth direction and information about a distance to the object in a depth direction; and deciding, based on the adjusted reliability, one of the detected subject/subjects as a main subject.

[0009] Further, according to the present invention, provided is an image processing method comprising: detecting one or more subjects from any of images obtained by repeatedly performing shooting; detecting a predetermined object from the image from which the subject/subjects are detected; determining one of the detected subject/subjects as a main subject based on a posture of each detected subject and a difference between information about a distance to each subject in the depth direction and information about a distance to the object in the depth direction.

[0010] Further, according to the present invention, provided is an image processing method comprising: detecting one or more subjects from any of images obtained by repeatedly performing shooting; detecting a predetermined object from the image from which the subject/subjects are detected; acquiring information regarding a distance in the depth direction to each detected subject detected and information regarding a distance in the depth direction to the object; determining a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject by using machine learning; and deciding one of the detected subject/subjects as a main subject based on the reliability, wherein training data optimized by using information about the distance to the main subject previously acquired is used in the machine learning.

[0011] Further, according to the present invention, provided is a non-transitory computer-readable storage medium, the storage medium storing a program that is executable by the computer, wherein the program includes program code for causing the computer to function as an electronic apparatus comprising: a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting; a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected; a determination unit that determines a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject detected by the first detection means; an adjustment unit that adjusts the reliability based on a difference between information about a distance to each of the subjects in a depth direction and information about a distance to the object in a depth direction; and a decision unit that decides, based on the reliability adjusted by the adjustment unit, one of the subject/subjects detected by the first detecting unit as a main subject.

[0012] Further, according to the present invention, provided is a non-transitory computer-readable storage medium, the storage medium storing a program that is executable by the computer, wherein the program includes program code for causing the computer to function as an electronic apparatus comprising: a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting; a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected; a decision unit that determines one of the subject/subjects detected by the first detection unit as a main subject based on a posture of each subject detected by the first detection unit and a difference between information about a distance to each subject in the depth direction and information about a distance to the object in the depth direction.

[0013] Further, according to the present invention, provided is a non-transitory computer-readable storage medium, the storage medium storing a program that is executable by the computer, wherein the program includes program code for causing the computer to function as an electronic apparatus comprising: a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting; a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected; an acquisition unit that acquires information regarding a distance in the depth direction to each subject detected by the first detection unit and information regarding a distance in the depth direction to the object; a determination unit that uses machine learning to determine a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject; and a decision unit that decides one of the subject/subjects detected by the first detection unit as a main subject based on the reliability, wherein the determination unit uses training data optimized by using information about the distance to the main subject previously acquired by the acquisition unit in the machine learning.

[0014] Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention, and together with the description, serve to explain the principles of the invention.

[0016] FIG. **1** is a block diagram illustrating a functional configuration of an image capturing apparatus according to an embodiment of the present invention.

[0017] FIG. **2** is a block diagram illustrating a functional configuration of a main subject determination unit according to a first embodiment.

[0018] FIG. **3** is a flowchart of main subject determination processing according to the first embodiment.

[0019] FIGS. **4**A and **4**B are conceptual diagrams of information acquired by a posture acquisition unit and an object detection unit according to the first embodiment.

[0020] FIG. **5** is a diagram illustrating an example of the structure of a neural network according to the first embodiment.

[0021] FIG. **6** is a diagram explaining an acquisition method in a depth information acquisition unit according to the first embodiment.

[0022] FIG. **7** is a conceptual diagram of information acquired by the depth information acquisition unit for each subject and by the depth information acquisition unit for an object according to the first embodiment.

[0023] FIG. **8** is a diagram illustrating a graph relating to an adjustment method of a reliability adjustment unit according to the first embodiment.

[0024] FIG. **9** is a block diagram illustrating a functional configuration of a main subject determination unit according to a second embodiment.

[0025] FIG. **10** is a flowchart of main subject determination processing according to the second embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0026] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention, and limitation is not made to an invention that requires a combination of all features described in the embodiments. Two or more of the multiple features described in the embodiments may be combined as appropriate. Furthermore, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

[0027] In the following embodiment, the present invention will be described as being implemented in an image capturing apparatus such as a digital camera, but the present invention can be applied to any electronic apparatus that can be equipped with a subject detection function and an image capturing function. Such electronic apparatuses include image capturing apparatuses, as well as computer devices (personal computers, tablet computers, media players, PDAs, etc.), mobile phones, smartphones, game consoles, robots, drones, drive recorders, etc. These are merely examples, and the present invention can be applied to other electronic apparatuses.

First Embodiment

Configuration of Image Capturing Apparatus

[0028] FIG. **1** is a block diagram illustrating an example of a functional configuration of an image capturing apparatus **100** according to the present embodiment. The image capturing apparatus **100** is capable of capturing and recording moving images and still images. The functional blocks in the image capturing apparatus **100** are connected to each other via a bus **160** in a communication-enabling manner. The operation of the image capturing apparatus **100** is realized by a main control unit **151**, which is composed of a central processing unit (CPU) or the like, executing a program and controlling each functional block.

[0029] An imaging lens **101** has a first fixed lens group **102**, a zoom lens **111**, an aperture **103**, a second fixed lens group **121**, a focus lens **131**, a zoom motor **112**, an aperture motor **104**, and a focus motor **132**. The first fixed lens group **102**, the zoom lens **111**, the aperture **103**, the second fixed lens group **121**, and the focus lens **131** constitute an imaging optical system. For illustrative convenience, the first fixed lens group **102**, the zoom lens **111**, the second fixed lens group **121**, and the focus lens **131** are each represented by a single lens, but each may be composed of a plurality of lenses. The imaging lens **101** may be configured as a detachable interchangeable lens.

[0030] An aperture control unit **105** controls the operation of the aperture motor **104** that actuates the aperture **103**, and changes the opening diameter of the aperture **103**. A zoom control unit **113** controls the operation of the zoom motor **112** that actuates the zoom lens **111**, and changes the focal length (angle of view) of the imaging lens **101**.

[0031] A focus control unit **133** calculates a defocus amount and defocus direction of the imaging lens **101** based on a phase difference between a pair of focus detection signals (image A and image B) obtained from an image sensor **141**. The focus control unit **133** then converts the defocus amount and defocus direction into an actuation amount and actuation direction of the focus motor **132**. Based on these actuation amount and actuation direction, the focus control unit **133** controls the operation of the focus motor **132** and actuates the focus lens **131** thereby controlling the focus state of the imaging lens **101**. In this way, the focus control unit **133** performs autofocus detection (AF) using a phase difference detection method.

[0032] Note that the AF method is not limited to the phase difference detection method. For example, the focus control unit **133** may perform AF using a contrast detection method based on a contrast evaluation value obtained from an image signal obtained from the image sensor **141**.

[0033] An image of a subject formed on the imaging plane of the image sensor **141** by the imaging lens **101** is converted into an electric signal (image signal) by photoelectric conversion elements in a plurality of pixels arranged in the image sensor **141**. In this embodiment, the image sensor **141** has pixels arranged in a matrix of m rows in the horizontal direction and n columns in the vertical direction (m and n are multiples), and each pixel is provided with a plurality of photoelectric conversion elements (photoelectric conversion regions) sharing one microlens. Note that here, it is assumed that two photoelectric conversion elements are provided, and a normal image can be obtained by adding the outputs (A signal and B signal) of these two photoelectric conversion elements. In addition, two images having parallax can be obtained by handling the outputs of the two photoelectric conversion elements individually. In the following description, a signal obtained by adding the outputs of the two photoelectric conversion elements is referred to as an image signal, and an image based on the image signal is simply referred to as an image. Further, the outputs of the two photoelectric conversion elements handled individually are referred to as a pair of parallax image signals, and the images based on the pair of parallax image signals are referred to as an A image and a B image, for distinction.

[0034] A shooting control unit **143** controls the reading of signals from the image sensor **141** in accordance with instructions from the main control unit **151**. Note that, from each pixel, the A signal and the B signal may be read out separately, or either the A signal or the B signal, and an A+B signal obtained by adding the A signal and the B signal, may be read out, as long as an image signal and a pair of parallax image signals are ultimately obtained.

[0035] The signal read out from the image sensor **141** is supplied to a signal processing unit **142**, which generates an image signal and a pair of parallax image signals. Then, in the signal processing unit **142**, signal processing such as noise reduction processing, A/D conversion processing, and automatic gain control processing is performed, and the resultant signals are output to the shooting control unit **143**. The shooting control unit **143** stores the image signal and the pair of parallax image signals received from the signal processing unit **142** in a random access memory (RAM) **154**.

[0036] An image processing unit **152** applies a predetermined image processing to the image signal stored in the RAM **154**. The image processing applied by the image processing unit **152** includes, but is not limited to, so-called development processing such as white balance adjustment processing, color interpolation (demosaic) processing, and gamma correction processing, as well as signal format conversion processing and scaling processing. In addition, the image processing unit **152** can also generate information on luminance of a subject for use in automatic exposure control (AE). Information on a specific subject area is supplied from a main subject determination unit **162** and may be used, for example, for white balance adjustment processing. In addition, in a case of performing AF using a contrast detection method, the image processing unit **152** may generate an AF evaluation value. The image processing unit **152** stores the processed image data in the RAM **154**.

[0037] In a case of recording image data stored in RAM **154**, main control unit **151** generates a data file according to the recording format by, for example, adding a predetermined header to the image data. At this time, the main control unit **151** controls a compression/decompression unit **153** to compress the amount of information by encoding the image data as necessary. The main control unit **151** records the generated data file on a recording medium **157**, such as a memory card.

[0038] Furthermore, in a case of displaying image data stored in the RAM **154**, the main control unit **151** controls the image processing unit **152** to scale the image data so as to fit the display size of a display unit **150**, and then writes the scaled image data to an area (VRAM area) of the RAM **154** used as a video memory. The display unit **150** reads out the image data for display from the VRAM area of the RAM **154**, and displays the image data on a display device such as an LCD or an organic EL display.

[0039] The image capturing apparatus **100** of this embodiment causes the display unit **150** to function as an electronic viewfinder (EVF) by instantly displaying on the display unit **150** an image of each frame captured during standby for still image shooting or during moving image recording. The moving image (frame images) displayed when the display unit **150** functions as an EVF are called a live view image or a through image. In addition, in a case where a still image is shot by the image capturing apparatus **100**, the image capturing apparatus **100** displays the shot still image on the display unit **150** for a certain period of time so that the user can check the shooting result. These display operations are also realized under the control of the main control unit **151**.

[0040] An operation unit **156** includes operation members such as switches, buttons, keys, a touch panel, and a line-of-sight input device that allow a user to input instructions to the image capturing apparatus **100**. An input through the operation unit **156** is detected by the main control unit **151** via the bus **160**, and the main control unit **151** controls each part to realize an operation according to the input.

[0041] The main control unit **151** has one or more programmable processors such as a CPU or an MPU, and controls each unit by, for example, loading programs stored in a storage unit **155** into a RAM **154** and executing them, thereby realizing the functions of the image capturing apparatus **100**. The main control unit **151** also executes AE processing to automatically determine exposure conditions (shutter speed or charge accumulation period, aperture value, and sensitivity) based on information about the luminance of the subject. The information about the luminance of the subject can be obtained, for example, from the image processing unit **152**. The main control unit **151** can also determine the exposure conditions based on a specific area of the subject, such as a person's face, as a reference.

[0042] Of the determined exposure conditions, the main control unit **151** notifies the shooting control unit **143** of the charge accumulation period and sensitivity (gain), and also notifies the aperture value to the aperture control unit **105**. The shooting control unit **143** controls the operation of the image sensor **141** so that shooting is performed according to the notified charge accumulation period, and sets the gain in the signal processing unit **142**. In addition, the aperture control unit **105** actuates the aperture motor **104** to control the opening diameter of the aperture **103** so as to be the notified aperture.

[0043] Batteries **159** are managed by a power management unit **158**, and supply power to the entire image capturing apparatus **100**.

[0044] A storage unit **155** stores the programs executed by the main control unit **151**, setting values required to execute the programs, GUI data, user setting values, weights and bias values learned by machine learning described below, etc. For example, when an instruction to transition from a power OFF state to a power ON state is given by operation of the operation unit **156**, the programs stored in the storage unit **155** are loaded into part of the RAM **154**, and the main control unit **151** executes the programs.

[0045] A depth information acquisition unit **161** calculates depth information from the pair of parallax image signals stored in the RAM **154**, and stores the calculated depth information in the RAM **154**.

[0046] The main subject determination unit **162** determines the main subject among the subjects detected in each captured frame image. The configuration and operation of the main subject determination unit **162** will be described in detail later. The result of determination by the main subject determination unit **162** is notified to each processing block via the bus **160**.

[0047] The result of determination by the main subject determination unit **162** can be used, for example, for automatic setting of a focus detection area. As a result, a tracking AF function for a specific subject area can be realized. In addition, AE processing can be performed based on the luminance information of the focus detection area, and image processing (e.g., gamma correction processing, white balance adjustment processing, etc.) can be performed based on the pixel values of the focus detection area. In addition, the main control unit **151** may superimpose an index (e.g., a rectangular frame surrounding an area) representing the current main subject or an area indicating a subject or unique object detected by the main subject determination unit **162** through processing described later on the displayed image.

Main Subject Determination Processing

[0048] Next, the main subject determination processing in this embodiment will be described with reference to FIGS. **2** and **3**.

[0049] FIG. **2** is a block diagram illustrating the functional configuration of the main subject determination unit **162**, and FIG. **3** is a flowchart illustrating the main subject determination processing. Note that, unless otherwise specified, each process in this flowchart is realized by each part of the image capturing apparatus **100** operating under the control of the main control unit **151**. Also, at the start of this flowchart, the image capturing apparatus **100** is powered on and in live view image shooting mode, and is in a state in which it is possible to instruct the start of shooting (recording) of a still image or moving image by operating the operation unit **156**. Note that, in the following explanation, a ball game played by a plurality of people is described as a scene to be shot in the main subject determination processing, but scenes to be shot to which this embodiment can be applied are not limited to this.

[0050] First, in step S**301**, the main control unit **151** controls the shooting control unit **143** to perform shooting by the image sensor **141**, and stores an image signal and a pair of parallax image signals A/D converted by the signal processing unit **142** in the RAM **154**.

[0051] In step S**302**, a subject detection unit **201** of the main subject determination unit **162** detects a subject (person) in the image of the image signal acquired in step S**301**. Any known method may be used as the subject detection method, and for example, the subject may be detected using a feature extraction process of a specific subject by Convolutional Neural Networks (CNN), or the subject to be detected may be registered in advance as a template and detected by template matching.

[0052] In step S**303**, a posture acquisition unit **202** estimates a posture of each of the subjects detected by the subject detection unit **201**, and acquires posture information. The contents of the posture information to be acquired are determined according to the type of the subject. In this example, since the subject is a person, the posture acquisition unit **202** acquires the positions of multiple joints of the person as posture information.

[0053] In step S**304**, an object detection unit **203** detects an object (hereinafter referred to as a "unique object") that is unique to the scene on the image of the image signal acquired in step **S301**, and acquires the two-dimensional coordinates and size of the detected unique object in the image. The type of unique object to be detected is determined based on the scene to be shot, and since the scene to be shot is a ball game in this example, the object detection unit **203** detects a ball as the unique object.

[0054] Any known method may be used for object detection and posture estimation. For example, the methods described in Reference 1 (Redmon, Joseph, et al., "You only look once: Unified, real-time object detection.", Proceedings of the IEEE conference on computer vision and pattern recognition, 2016) and Reference 2 (Cao, Zhe, et al., "Real-time multi-person 2d pose estimation using part affinity fields.", Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017) may be used.

[0055] FIGS. **4**A and **4**B are conceptual diagrams of information acquired by the posture acquisition unit **202** and the object detection unit **203**. FIG. **4**A shows an example of an image to be processed, in which a subject **402** is about to kick a ball **403** and a subject **401** is about to interfere with the subject **402**'s kicking of the ball **403**. The important subject in this scene is the subject **402** about to kick the ball. In this embodiment, the posture information of the subject and the information of the unique object are used to determine a main subject that is likely to be intended by the user as a target for image shooting control. On the other hand, the subject **401** is a non-main subject. Here, the non-main subject refers to a subject other than the main subject.

[0056] FIG. **4**B is a diagram showing an example of posture information of the subject **401** and the subject **402**, and the position and size of the ball **403**. Joints **411** represent respective joints of the subject **401**, and joints **412** represent respective joints of the subject **402**. FIG. **4**B shows an example in which the positions of the top of the head, the neck, the shoulders, the elbows, the wrists, the hip joints, the knees, and the ankles are acquired as joints, but the acquired joints may be some of these, or positions other than the joints may be acquired. In addition, the posture information may be information on axes connecting the joints, etc., in addition to the joint positions, and any information may be used as long as it represents the posture of the subject. In the following description, a case in which the joint positions are acquired as posture information will be described.

[0057] The posture acquisition unit **202** acquires the two-dimensional coordinates (x, y) of the joints **411** and **412** in the image. Here, the unit of (x, y) is a pixel. A center of gravity position **413** represents the center of gravity position of the ball **403**, and an arrow **414** represents the size of the ball **403** in the image. The object detection unit **203** acquires the two-dimensional coordinates (x, y) of the center of gravity position of the ball **403** in the image, and the number of pixels indicating the diameter which is the size of the ball **403** in the image.

[0058] In step **S305**, a reliability calculation unit **204** calculates a reliability (probability) representing the likelihood of each subject being a main subject based on at least one of the coordinates of the joints acquired by the posture acquisition unit **202** and the coordinates and size of the unique object acquired by the object detection unit **203**. This reliability corresponds to the degree of possibility that the subject is the main subject of the image to be processed. A method for calculating the reliability will be described later. In this embodiment, a case will be described in which the probability that the subject is the main subject of the image to be processed is used as the reliability representing the likelihood of the subject being a main subject, but a value other than the probability may be used. For example, the reciprocal of the distance between the center of gravity of the subject and the center of gravity of the unique object may be used as the reliability.

[0059] Here, a method for calculating the probability that represents the likelihood of a subject being a main subject based on the coordinates of each joint and the coordinates and size of a unique object, which is performed in the reliability calculation unit **204**, will be described. In the following description, a case where a neural network, which is one method of machine learning, is used will be described.

[0060] FIG. **5** is a diagram showing an example of the structure of a neural network. In FIG. **5**, reference numeral **501** indicates an input layer; **502**, an intermediate layer; **503**, an output layer; **504**, neurons; and **505**, the connection relationship between the neurons **504**. Here, for convenience of illustration, only representative neurons and connection lines are numbered. The number of neurons **504** in the input layer **501** is equal to the dimension of the input data, and the number of neurons in the output layer **503** is two. This corresponds to a two-class classification problem of determining whether or not the data represents the main subject.

[0061] A weight wji is assigned to a line **505** connecting the i-th neuron **504** in the input layer **501** and the j-th neuron **504** in the intermediate layer **502**, and the value z_j output by the j-th neuron **504** in the intermediate layer **502** can be expressed by the following Equations (1) and (2).

$$z_j = h\left(b_j + \sum\nolimits_j w_{ji} x_i\right) \tag{1}$$

$$h(z) = \max(z, 0) \tag{2}$$

[0062] In the Equation (1), x_i represents the value input to the i-th neuron **504** in the input layer **501**. The sum is taken for all neurons **504** in the input layer **501** that are connected to the j-th neuron. b_j is called the bias, and is a parameter that controls the ease with which the j-th neuron **504** fires. The function h defined by the Equation (2) is an activation function called Rectified Linear Unit (ReLU). As the activation function, it is also possible to use another function such as a sigmoid function.

[0063] Moreover, a value y_k output by the k-th neuron **504** in the output layer **503** is given by the following Equations (3) and (4).

$$y_k = f\left(b_k + \sum\nolimits_j w_{kj} z_j\right) \tag{3}$$

$$f(y_k) = \frac{\exp(y_k)}{\sum_i \exp(y_i)} \tag{4}$$

[0064] In the Equation (3), z_j represents the value output by the j-th neuron **504** in the intermediate layer **502**, where i, k=0, 1. 0 corresponds to a non-main subject, and 1 corresponds to a main subject. The sum is taken for all neurons in the intermediate layer **502** that are connected to the k-th neuron. Furthermore, the function f defined in the Equation (4) is called a softmax function, and outputs a probability value that belongs to the k-th class. In this embodiment, $f(y_1)$ is used as the probability representing the likelihood of a subject being a main subject.

[0065] During learning, the coordinates of the joints of the subject and the coordinates and size of the unique object are input. Then, all weights and biases are optimized so as to minimize a loss function that uses the output probability and the correct label. Here, the correct label takes two values: "1" for the main subject and "0" for a non-main subject. As the loss function L, a binary cross-entropy as expressed by the following Equation (5) may be used.

$$L(y, t) = -\sum_m t_m \log y_m - \sum_m (1 - t_m)\log(1 - y_m) \tag{5}$$

[0066] In the Equation (5), the subscript m represents the index of the subject to be learned. y_m is the probability value output from the neuron **504** with k=1 in the output layer **503**, and t_m is the correct label. The loss function may be any function other than the Equation (5) that can measure the degree of agreement with the correct label, such as the mean square error function. By optimizing the weights and biases based on the Equation (5), it is possible to determine the weights and biases so that the correct label and the output probability value approach each other.

[0067] The learned weights and bias values (learned data) are saved in advance in the storage unit **155**, and are stored in the RAM **154** as necessary. A plurality of types of weights and bias values may be prepared depending on the scene. The reliability calculation unit **204** uses the learned weights and biases (the results of machine learning performed in advance) to output the probability value $f(y_1)$ based on the Equations (1) to (4).

[0068] In addition, when learning, the state before moving on to an important action may be learned as the state of the main subject. For example, in the case of kicking a ball, the state of swinging the leg up to kick the ball can be learned as one of the states of the main subject. The reason for adopting this configuration is that the control of the image capturing apparatus **100** needs to be executed appropriately when the main subject actually performs an important action. For example, when the reliability (probability) corresponding to the main subject exceeds a preset threshold, the control to automatically record a still image or a moving image (recording control) is started, so that the user can capture an image of an important moment without missing it. In this case, information on the typical time taken from the state of the learned target to the important action may be used for the control of the image capturing apparatus **100**.

[0069] Although the method for calculating the reliability (probability) using a neural network has been described above, other machine learning methods such as a support vector machine or a decision tree may be used as long as it is possible to classify whether a subject is a main subject or not. Also, without being limited to machine learning, a function that outputs the reliability or probability based on a certain model may be constructed.

[0070] In the scene shown in FIG. **4**A, it is assumed that both subjects **401** and **402** are in a position with their leg swung up, there is no significant difference in their coordinate distance from the ball **403**, and there is no significant difference in their detected reliability.

[0071] In step S**306**, the depth information acquisition unit **161** generates depth information using the pair of parallax image signals. The depth information is one piece of information about the distance in the depth direction of each pixel with reference to the position of the image capturing apparatus **100**, and is also called a distance map, a depth map, a depth image, or a distance image. Note that the depth information may be generated without using the pair of parallax image signals. For example, the position of the focus lens **131** where the contrast evaluation value is maximized may be obtained for each pixel to generate depth information for each pixel.

[0072] Here, using FIG. **6**, a method for calculating distance information to a subject at each pixel will be explained as an example of depth information.

[0073] If an image A **601** and image B **602** are obtained, it can be seen that the light beam is refracted as shown by the solid line from the focal length of the imaging lens **101** and the distance information between the focus lens **131** and the image sensor **141**. It can therefore be seen that the subject in focus is at position **605**. Similarly, if the image A **601** and an image B **603** are obtained, it can be seen that the subject in focus is at position **606**, and if the image A **601** and an image B **604** are obtained, it can be seen that the subject in focus is at position **607**. As described above, for each pixel, distance information to the subject at that pixel position can be calculated from the relative position between an image A which includes that pixel and a corresponding image B.

[0074] For example, in FIG. **6**, it is assumed that the image A **601** and the image B **604** are obtained. In this case, a distance **608** from a pixel **609** located at the midpoint, which corresponds to half the shift amount between the images, to a position **607**, or a defocus amount corresponding to the distance **608**, is stored as distance information for the pixel **609**. In this manner, distance information can be obtained for each pixel, and depth information can be generated.

[0075] The depth information may be generated by dividing the image into minute regions and calculating the defocus amount for each minute region. In that case, images A and B are generated from pixels included in each minute region, and the phase difference (image shift amount) is detected by correlation calculation and converted into the defocus amount. In this case, too, the generated depth information indicates the distance information of each pixel, but pixels included in the same minute region have the same distance information. The depth information acquisition unit **161** supplies the generated depth information to a subject depth information acquisition unit **205** and an object depth information acquisition unit **206**.

[0076] In the above description, the case where the defocus amount, which is distance information of subjects and object, is acquired as the depth information has been described in detail, but the present invention is not limited to this. As described above, the depth information may be a distance map, a depth map, a depth image, or a distance image. Further, for example, values that the defocus amounts atr normalized by the focal depth (e.g., 1Fδ, where F is the aperture value and δ is a diameter of the allowable circle of confusion) may be obtained as the depth information. Furthermore, the depth information may be, for example, an actual distance indicating the actual distance from the image capturing apparatus **100** to the subject, or information indicating the image shift amount used to derive the defocus amount.

[0077] The processes of steps S**302** and S**303**, S**304**, and S**306** may be performed in parallel.

[0078] Next, in step S**307**, the subject depth information acquisition unit **205** acquires depth information of each subject detected by the subject detection unit **201** using the depth information acquired by the depth information acquisition unit **161**.

[0079] Furthermore, in step S**308**, the object depth information acquisition unit **206** acquires depth information of the unique object detected by the object detection unit **203** using the depth information acquired by the depth information acquisition unit **161**.

[0080] Note that the processes of steps S**307** and S**308** may be performed in parallel.

[0081] FIG. **7** is a conceptual diagram showing the results of applying processing by the subject depth information acquisition unit **205** and the object depth information acquisition unit **206** to the subjects **401** and **402** and the ball **403** in FIG. **4**A. The diagonal lines and shading in FIG. **7** each represent distance information, with the diagonal lines indicating closer to the image capturing apparatus **100**. The subject **402** has the same depth as the ball **403**, and the subject **401** is further back (farther away from the image capturing apparatus **100**) than the subjects **402** and the ball **403**.

[0082] In step S**309**, a reliability adjustment unit **207** adjusts the reliabilities calculated by the reliability calculation unit **204** using the depth information acquired by the subject depth information acquisition unit **205** and the object depth information acquisition unit **206**. For example, the reliabilities (probabilities) $f(y_1)$ obtained using the Equations (1) to (4) are adjusted by multiplying them by a coefficient k corresponding to the absolute values of the differences between the distance information of the subjects and the distance information of the object.

[0083] FIG. **8** is a graph showing the relationship between the coefficient (amount of change) and the absolute value of the difference between the distance information of a subject and the distance information of a unique object. As shown in FIG. **8**, the coefficient takes a value between 0 and 1, and when the difference is 0, the coefficient k is set to 1, and as the difference increases, the coefficient k approaches 0, thereby decreasing the reliability (probability).

[0084] The slope of the graph may also be changed according to the distance between a unique object and a subject on the image. d(n) is the absolute value of the difference in distance information between the subject and the unique object in which the coefficient k is 0 when the distance between the subject and the unique object in the image is n, and the straight line **801** is the graph under such condition. Similarly, d(m) is the absolute value of the difference in distance information between the subject and the unique object in which the coefficient k is 0 when the distance between the subject and the unique object in the image is m, and the straight line **802** is the graph under such condition. In this embodiment, as the distance between the subject and the object in the image becomes shorter, the absolute value of the difference in distance information between the subject and the unique object when the coefficient k is 0 is increased to make the slope of the coefficient k gentler. In the example shown in FIG. **8**, the distance is $m<n$.

[0085] Moreover, in a case where the reliability $f(y_1)$ is equal to or greater than a predetermined threshold, the coefficient k may not be multiplied. Moreover, the threshold may be adjusted according to the absolute value of the difference in the distance information between the subject and the object.

[0086] In the present embodiment, the relationship between the coefficient and the absolute value of the difference in distance information between the subject and the object is described as a linear function (straight line), but it may be a logarithmic function or an exponential function. Also, in the present embodiment, the minimum value of the coefficient k is set to 0, but the minimum value may be set to a value greater than 0.

[0087] In the example shown in FIG. **7**, the absolute value of the difference in distance information between the subject **402** and the ball **403** is smaller than that between the subject **401** and the ball **403**, and therefore as a result of the reliability adjustment, the rate of decrease in the reliability of the subject **402** as the main subject is lower than the rate of decrease in the reliability of the subject **401** as the main subject.

[0088] In step S**310**, a main subject determination unit **208** determines the main subject of the image acquired in step S**301** based on the reliabilities adjusted by the reliability adjustment unit **207**, stores the main subject information of the determined main subject in the RAM **154**, and ends the processing.

[0089] In a scene shown in FIG. **4**A, it is assumed that the subject with the highest reliability is the subject **402** among the subjects whose reliabilities have been adjusted by the reliability adjustment unit **207**. In this case, if the current main subject stored in the RAM **154** is the subject **401**, the main subject is changed from the subject **401** to the subject **402** having the highest reliability.

[0090] In a case of changing the main subject, hysteresis may be provided, such as changing the main subject from the subject **401** to the subject **402** if the subject **402** has the highest reliability a predetermined number of times over time. In this case, the timing to change to the subject **402** is determined using past main subject information stored in the RAM **154**. Also, the main subject may be changed if the distance in the coordinates between the current main subject and the subject with the highest reliability in the current frame image is constant. Also, the main subject may be changed using any chronological information.

[0091] As described above, according to the first embodiment, the image capturing apparatus **100** acquires posture information of each of the plurality of subjects detected from the image to be processed, a unique object, and depth information. The image capturing apparatus **100** then calculates a reliability, corresponding to the degree of likelihood that the subject is a main subject of the image to be processed, of each of the plurality of subjects based on the posture information and the unique object. The image capturing apparatus **100** then adjusts the calculated reliabilities using the depth information, thereby determining a main subject of the image to be processed from among the plurality of subjects based on the plurality of adjusted reliabilities of the plurality of subjects. This makes it possible to determine a main subject that is likely to meet the user's intention in an image containing a plurality of subjects.

[0092] In the above example, the main subject is determined for each frame image, but the determination may be performed every predetermined number of frame images.

Second Embodiment

[0093] Next, a second embodiment of the present invention will be described. Note that the same reference numerals will be used for configurations and processes common to the first embodiment, and descriptions thereof will be omitted.

[0094] The second embodiment will be described with reference to FIGS. **9** and **10**. FIG. **9** is a block diagram illustrating a functional configuration of main subject determination unit **162** in the second embodiment, and FIG. **10** is a flowchart of main subject determination processing in the second embodiment.

[0095] After acquiring depth information by the depth information acquisition unit **161** in step S**306**, in step S**1001**, a reliability calculation unit **901** calculates a reliability (probability) representing the likelihood of each subject being a main subject based on the coordinates of the joints estimated by the posture acquisition unit **202**, and at least one of the coordinates and size of the unique object acquired by the object detection unit **203**, and the depth information acquired by the depth information acquisition unit **161**. The method of calculating the probability is the same as that described in the first embodiment, and therefore a description thereof will be omitted.

[0096] In the first embodiment, the coordinates of the joints of the subject and the coordinates and size of the unique object are used during learning, and the loss function L is optimized using based on the Equation (5). In contrast, in the second embodiment, depth information acquired by the depth information acquisition unit **161** is additionally input, and the loss function L is optimized based on the Equation (5).

[0097] In step S**1002**, a main subject determination unit **902** determines the main subject of the image acquired in step S**301** based on the reliability calculated by the reliability calculation unit **901**. Note that the main subject determination method is similar to that described in the first embodiment, and therefore description thereof will be omitted.

[0098] As described above, according to the second embodiment, by using depth information as an input of learning in advance, the image capturing apparatus **100** calculates a reliability, corresponding to the degree of likelihood that the subject is a main subject, for each of a plurality of subjects based on the posture information, unique object, and depth information. Then, based on the calculated plurality of reliabilities, the image capturing apparatus **100** determines a main subject from among the plurality of subjects in the image to be processed. This makes it possible to determine a main subject that is likely to meet the user's intention in an image containing a plurality of subjects.

<Modification>

[0099] In the first and second embodiments described above, it has been explained that posture information of the detected subject is obtained without error and that an appropriate reliability is obtained. However, if the movement of the main subject intended by the photographer is large and fast, there is a possibility that posture information acquisition will fail and an appropriate reliability will not be obtained.

[0100] For example, in the scene shown in FIG. **4**A, if the movement of the subject **402** is fast and posture information acquisition fails, the reliability of the subject **402** will be lower than that of the subject **401**.

[0101] In this modification, as a process to deal with such a case, if the subject **402** is stored as the main subject in the history of main subjects stored in the RAM **154** and the subject **401** is selected as the main subject based on the reliability, the following determination is made. That is, the absolute value of the difference in the distance information between the subject **401** and the ball **403** is compared with a threshold. Then, if the absolute value of the difference is equal to or greater than the threshold, the subject **402** is maintained as the main subject regardless of the reliability, and if it is smaller than the threshold, the main subject is changed to the subject **401**.

[0102] By controlling in this way, even if acquisition of posture information fails, frequent changes of the main subject can be prevented.

OTHER EMBODIMENTS

[0103] The present invention may be applied to a system made up of a plurality of devices, or to an apparatus made up of a single device.

[0104] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0105] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0106] This application claims the benefit of Japanese Patent Application No. 2024-027866, filed Feb. 27, 2024 which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An electronic apparatus comprising one or more processors and/or circuitry which function as:

a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting;

a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected;

a determination unit that determines a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject detected by the first detection unit;

an adjustment unit that adjusts the reliability based on a difference between information about a distance to each of the subjects in a depth direction and information about a distance to the object in a depth direction; and

a decision unit that decides, based on the reliability adjusted by the adjustment unit, one of the subject/subjects detected by the first detecting unit as a main subject.

2. The electronic apparatus according to claim **1**, wherein the adjustment unit reduces a decrease in the reliability in a case where the difference is a first value, compared to a case where the difference is a second value that is greater than the first value.

3. The electronic apparatus according to claim **2**, wherein the adjustment unit reduces a rate of decrease according to the difference in a case where a distance in the image between each of the subjects and the object is a first distance, compared to a case where the distance in the image is a second distance that is greater than the first distance.

4. The electronic apparatus according to claim **1**, wherein the adjustment unit does not adjust the reliability in a case where the reliability is equal to or greater than a predetermined threshold.

5. The electronic apparatus according to claim **1**, wherein the adjustment unit adjusts the reliability by multiplying the reliability by a coefficient determined based on the difference, and

the coefficient is equal to or greater than 0 and equal to or less than 1.

6. The electronic apparatus according to claim **5**, wherein the coefficient is defined by any one of a linear function, a logarithmic function, and an exponential function of the difference.

7. The electronic apparatus according to claim **1**, wherein the adjustment unit obtains information about a distance in a depth direction for each pixel of the image, and obtains information about a distance to each of the subjects and a distance to the object based on the obtained information about the distance for each pixel.

8. The electronic apparatus according to claim **1**, wherein the adjustment unit obtains information about a distance in a depth direction for each of a plurality of regions obtained by dividing the image, and obtains information about the distance to each of the subjects and a distance to the object based on the obtained information about the distance for each region.

9. The electronic apparatus according to claim **1**, wherein the information about the distance includes a defocus amount, a value obtained by normalizing the defocus amount by a focal depth, a distance from the electronic apparatus, and an image shift amount.

10. The electronic apparatus according to claim **1**, wherein the determination unit determines the reliability by machine learning.

11. The electronic apparatus according to claim **10**, wherein the machine learning includes a neural network that is trained on the subject, a support vector machine, and a decision tree.

12. The electronic apparatus according to claim **1**, wherein the one or more processors and/or circuitry further function as a storage unit that stores the main subject determined by the determination unit, and

the decision unit selects one of the subjects detected by the first detection unit as a main subject based on the reliability, and determines the main subject based on the selected main subject and a history of main subjects stored in the storage unit.

13. The electronic apparatus according to claim **12**, wherein, in a case where a main subject stored in the storage unit is detected by the first detection unit and the stored main subject is different from the selected main subject, the decision unit decides the main subject stored in the storage unit as the main subject if a difference in information about a distance in a depth direction between the selected main subject and the object is greater than a difference in a distance in a depth direction between the main subject stored in the storage unit and the object.

14. The electronic apparatus according to claim **12**, wherein, in a case where a main subject stored in the storage unit is detected by the first detection unit and the stored main subject is different from the selected main subject, the decision unit determines the selected subject as the main subject if the same subject is selected as the main subject a predetermined number of times in succession.

15. The electronic apparatus according to claim **1**, wherein the predetermined object is determined according to a scene to be shot.

16. The electronic apparatus according to claim **1** further comprising an image shooting unit that shoots the images.

17. An electronic apparatus comprising one or more processors and/or circuitry which function as:

a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting;

a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected;

a decision unit that decides one of the subject/subjects detected by the first detection unit as a main subject based on a posture of each subject detected by the first detection unit and a difference between information about a distance to each subject in the depth direction and information about a distance to the object in the depth direction.

18. The electronic apparatus according to claim **17**, wherein the information about the distances to each subject and the object is obtained, by obtaining information about the distance in the depth direction for each pixel of the image, based on the obtained information about the distance for each pixel.

19. The electronic apparatus according to claim **17**, wherein the information about the distances to each subject and the object is obtained, by obtaining information about the distance in the depth direction for each of a plurality of regions obtained by dividing the image, based on the obtained information about the distance for each region.

20. The electronic apparatus according to claim **17**, wherein the information about the distance includes a defocus amount, a value obtained by normalizing the defocus amount by a focal depth, a distance from the electronic apparatus, and an image shift amount.

21. The electronic apparatus according to claim **17**, wherein the predetermined object is determined according to a scene to be shot.

22. The electronic apparatus according to claim **17** further comprising an image shooting unit that shoots the images.

23. An electronic apparatus comprising one or more processors and/or circuitry which function as:

a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting;

a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected;

an acquisition unit that acquires information regarding a distance in the depth direction to each subject detected by the first detection unit and information about a distance in the depth direction to the object;

a determination unit that uses machine learning to determine a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject; and

a decision unit that decides one of the subject/subjects detected by the first detection unit as a main subject based on the reliability,

wherein the determination unit uses training data optimized by using information about the distance to the main subject previously acquired by the acquisition unit in the machine learning.

24. The electronic apparatus according to claim **23**, wherein the one or more processors and/or circuitry further functions as a learning unit that optimizes the training data to be used in the machine learning by using the information about the distance acquired by the acquisition unit.

25. The electronic apparatus according to claim **23**, wherein the one or more processors and/or circuitry further functions as a storage unit that stores the main subject decided by the decision unit,

wherein the decision unit selects one of the subject/subjects detected by the first detection unit as a main subject based on the reliability, and decides the main subject based on the selected main subject and a history of main subjects stored in the storage unit.

26. The electronic apparatus according to claim **25**, wherein, in a case where a main subject stored in the storage unit is detected by the first detection unit and the stored main subject is different from the selected main subject, the decision unit decides the main subject stored in the storage unit as the main subject if a difference in information about a distance in a depth direction between the selected main subject and the object is greater than a difference in a distance in a depth direction between the main subject stored in the storage unit and the object.

27. The electronic apparatus according to claim **25**, wherein, in a case where a main subject stored in the storage unit is detected by the first detection unit and the stored main subject is different from the selected main subject, the decision unit decides the selected subject as the main subject if the same subject is selected as the main subject a predetermined number of times in succession.

28. The electronic apparatus according to claim **23**, wherein the predetermined object is determined according to a scene to be shot.

29. The electronic apparatus according to claim **23** further comprising an image shooting unit that shoots the images.

30. An image processing method comprising:

detecting one or more subjects from any of images obtained by repeatedly performing shooting;

detecting a predetermined object from the image from which the subject/subjects are detected;

determining a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each detected subject;

adjusting the reliability based on a difference between information about a distance to each of the subjects in a depth direction and information about a distance to the object in a depth direction; and

deciding, based on the adjusted reliability, one of the detected subject/subjects as a main subject.

31. An image processing method comprising:

detecting one or more subjects from any of images obtained by repeatedly performing shooting;

detecting a predetermined object from the image from which the subject/subjects are detected;

determining one of the detected subject/subjects as a main subject based on a posture of each detected subject and a difference between information about a distance to each subject in the depth direction and information about a distance to the object in the depth direction.

32. An image processing method comprising:

detecting one or more subjects from any of images obtained by repeatedly performing shooting;

detecting a predetermined object from the image from which the subject/subjects are detected;

acquiring information regarding a distance in the depth direction to each detected subject detected and information regarding a distance in the depth direction to the object;

determining a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject by using machine learning; and

deciding one of the detected subject/subjects as a main subject based on the reliability,

wherein training data optimized by using information about the distance to the main subject previously acquired is used in the machine learning.

33. A non-transitory computer-readable storage medium, the storage medium storing a program that is executable by the computer, wherein the program includes program code for causing the computer to function as an electronic apparatus comprising:

a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting;

a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected;

a determination unit that determines a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject detected by the first detection means;

an adjustment unit that adjusts the reliability based on a difference between information about a distance to each of the subjects in a depth direction and information about a distance to the object in a depth direction; and

a decision unit that decides, based on the reliability adjusted by the adjustment unit, one of the subject/subjects detected by the first detecting unit as a main subject.

34. A non-transitory computer-readable storage medium, the storage medium storing a program that is executable by the computer, wherein the program includes program code for causing the computer to function as an electronic apparatus comprising:

a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting;

a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected;

a decision unit that determines one of the subject/subjects detected by the first detection unit as a main subject based on a posture of each subject detected by the first detection unit and a difference between information about a distance to each subject in the depth direction and information about a distance to the object in the depth direction.

35. A non-transitory computer-readable storage medium, the storage medium storing a program that is executable by the computer, wherein the program includes program code for causing the computer to function as an electronic apparatus comprising:

a first detection unit that detects one or more subjects from any of images obtained by repeatedly performing shooting;

a second detection unit that detects a predetermined object from the image from which the subject/subjects are detected;

an acquisition unit that acquires information regarding a distance in the depth direction to each subject detected by the first detection unit and information regarding a distance in the depth direction to the object;

a determination unit that uses machine learning to determine a reliability indicating a degree of possibility of a subject being a main subject based on a posture of each subject; and

a decision unit that decides one of the subject/subjects detected by the first detection unit as a main subject based on the reliability,

wherein the determination unit uses training data optimized by using information about the distance to the main subject previously acquired by the acquisition unit in the machine learning.

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